

Exploratory Data Analysis (EDA) on Housing Dataset

Introduction

Exploratory Data Analysis (EDA) is an important step in data science that helps understand data characteristics, identify patterns, detect anomalies, and discover relationships between variables.

This project analyzes a Housing dataset to identify the factors influencing house prices and uncover meaningful insights.

Objective

- Understand the structure of the dataset.
 - Analyze the distribution of house prices.
 - Identify relationships between housing features and price.
 - Determine the most influential factors affecting house prices.
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Dataset Overview

- Dataset Name: Housing Dataset
- Total Records: 545
- Features: 13
- Target Variable: Price

The dataset contains information about residential properties including area, bedrooms, bathrooms, stories, parking, furnishing status, and other amenities.

Data Inspection

Missing Values Analysis

The dataset was checked for missing values.

Observation

- No missing values were found.
- Dataset is clean and ready for analysis.

Statistical Summary

Descriptive statistics were generated for all numerical variables.

Key Observations

- House prices vary significantly across properties.
 - Property area shows a wide range of values.
 - Most houses contain between 2 and 4 bedrooms.
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Visualizations

1. House Price Distribution

A histogram was used to analyze the distribution of house prices.

Observation

- Most houses are concentrated in the medium price range.
 - High-priced houses are fewer in number.
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2. Correlation Heatmap

A correlation matrix was created to identify relationships between variables.

Observation

- Area shows a strong positive relationship with price.
 - Bathrooms and stories also influence house prices.
 - Several amenities positively contribute to property value.
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3. Area vs House Price

A scatter plot was used to study the relationship between property area and price.

Observation

- Larger houses generally have higher prices.
- Positive correlation exists between area and house price.

4. Bedrooms vs House Price

A boxplot was used to compare house prices across bedroom counts.

Observation

- Houses with more bedrooms tend to have higher prices.
- Price variability increases with the number of bedrooms.

Key Findings

- Area is one of the strongest predictors of house price.
- Bathrooms significantly affect property value.
- Additional facilities such as air conditioning and preferred area contribute to higher prices.
- The dataset is clean and suitable for predictive modeling.

Conclusion

The Exploratory Data Analysis successfully identified the major factors affecting house prices. Area, bathrooms, and property amenities were found to have strong relationships with house price. The dataset contains no missing values and provides a solid foundation for machine learning and predictive analytics.

The objectives of the EDA project were successfully achieved.